

Data Mining Project Report

Title

CFPB Consumer Complaints: Predicting Monetary Relief Outcomes from
Complaint-Time Information

Names

Amin Ben Abdelhafidh
Koussay Hidouri
Louay Ilahi

Course Title

BA 360: Business Data Mining

Contents

1	Dataset Description	1
1.1	General Context	1
1.2	Data Source	1
1.3	Unit of Observation	1
1.4	Main Variables	1
2	Problem Statement	2
2.1	Problem to Study	2
2.2	Objective	2
2.3	Research Questions	2
2.4	Target Variable	2
3	Data Preparation	3
3.1	Missing Values	3
3.2	Outliers and Quality Checks	3
3.3	Feature Transformations	3
3.4	Encoding and Scaling	3
3.5	Feature Selection and Leakage Control	3
4	Methods and Analysis	3
4.1	Modeling Strategy	3
4.2	Models Implemented	3
4.3	Why These Methods	3
5	Results and Interpretation	3
5.1	Model Comparison	3
5.2	Threshold Analysis	3
5.3	Confusion Matrix Insights	5
5.4	Key Findings	5
6	Conclusion	5
6.1	Objective Recap	5
6.2	Methods Recap	5
6.3	Main Results	5
6.4	Business Insights	5
7	Appendix (Optional)	5
7.1	Evolution and Corrections	5
7.2	Additional Tables	5

1 Dataset Description

1.1 General Context

The dataset comes from the U.S. Consumer Financial Protection Bureau (CFPB). It contains real consumer complaints about financial products and services (for example credit cards, loans, credit reporting, and debt collection). In business terms, this dataset helps identify where customer harm occurs and where monetary relief is more likely, which supports better risk monitoring and faster case prioritization.

1.2 Data Source

- Source: CFPB Consumer Complaint Database.
- Official page: <https://www.consumerfinance.gov/data-research/consumer-complaints/>
- Direct file used as full raw input in this project: `data/raw/complaints.csv`.
- Because the full file is very large, we created and used a memory-safe working sample for modeling: `data/processed/complaints_sample.csv`.

Local snapshot (as documented in the project README on 2026-04-20):

- Full raw dataset size: 14,636,145 rows and 18 columns.
- Uncompressed CSV size: about 8.0 GB.
- Compressed download size (`complaints.csv.zip`): about 1.7 GB.

1.3 Unit of Observation

One observation (one row) represents one consumer complaint submitted to the CFPB complaint system. Each row includes complaint metadata available at intake time (for example product, issue, submission channel, state, and dates), and in many cases a free-text complaint narrative.

For this project, we defined a binary target from resolution information:

- 1: complaint ended with monetary relief.
- 0: complaint ended without monetary relief.

To avoid leakage, post-resolution fields are not used as predictors at inference time.

1.4 Main Variables

Main variables used in EDA and modeling include:

- Text variable: `consumer_complaint_narrative`.
- Product taxonomy: `product` and `sub_product`.
- Problem taxonomy: `issue` and `sub_issue`.
- Submission and geography: `submitted_via` and `state`.
- Time variables: `date_received` (and derived temporal features).
- Target construction fields: `company_response_to_consumer` (used to derive the label, then excluded from predictor set for leakage-safe modeling).

Data handling note: because the full dataset in `data/raw/complaints.csv` is too large for efficient iterative experimentation on a local machine, we generated a representative sample

(data/processed/complaints_sample.csv) and used that sample throughout preprocessing and model development.

2 Problem Statement

2.1 Problem to Study

Financial institutions and regulators receive a very large volume of consumer complaints, but only a subset of cases ends with monetary relief for the consumer. The core problem is to identify, as early as possible, which complaints are more likely to lead to monetary relief using only information available at complaint time.

This is a pressing business problem for companies because slow or inaccurate complaint triage increases regulatory exposure, operational costs, customer dissatisfaction, and reputational risk. In practice, firms must quickly decide which complaints need urgent escalation versus standard handling. A reliable predictive signal can improve response prioritization, shorten time-to-resolution for high-risk cases, and support more consistent consumer-outcome management.

2.2 Objective

The objective of this analysis is to build and evaluate a leakage-safe binary classification model that predicts whether a CFPB complaint will end with monetary relief. We aim to compare multiple modeling approaches, assess class-imbalance-aware performance metrics, and identify a practical decision threshold for operational use.

2.3 Research Questions

The main research questions are:

- Can complaint-time features (text narrative, product/issue categories, channel, geography, and time features) predict monetary relief outcomes with useful accuracy?
- Which model family provides the best trade-off between precision, recall, and overall robustness for this imbalanced classification task?
- How should the decision threshold be adjusted to match operational priorities, such as capturing more potential monetary-relief cases versus reducing false alarms?
- Which complaint attributes appear most informative for distinguishing likely monetary-relief outcomes?

2.4 Target Variable

The target variable is a binary label derived from the complaint resolution outcome:

- 1: complaint ended with monetary relief.
- 0: complaint ended without monetary relief.

To preserve real-world validity, predictors are restricted to complaint-time information only. Fields that reveal or are strongly tied to post-resolution outcomes are excluded from the feature set used at inference time.

3 Data Preparation

3.1 Missing Values

Explain handling strategy and justification.

3.2 Outliers and Quality Checks

Describe checks and decisions.

3.3 Feature Transformations

Detail text cleaning, vectorization, and any transformations.

3.4 Encoding and Scaling

Document categorical encoding and scaling choices.

3.5 Feature Selection and Leakage Control

List excluded post-resolution fields and explain why.

4 Methods and Analysis

4.1 Modeling Strategy

Explain train/test setup and model comparison approach.

4.2 Models Implemented

- Logistic Regression (TF-IDF)
- Naive Bayes (TF-IDF)
- KNN (SVD-reduced text)
- Random Forest (structured + text)
- MLP Neural Network
- Voting Classifier Ensemble

4.3 Why These Methods

Justify method choices for this problem.

5 Results and Interpretation

5.1 Model Comparison

Insert summary table from reports outputs.

5.2 Threshold Analysis

Discuss trade-offs and operational threshold recommendation.

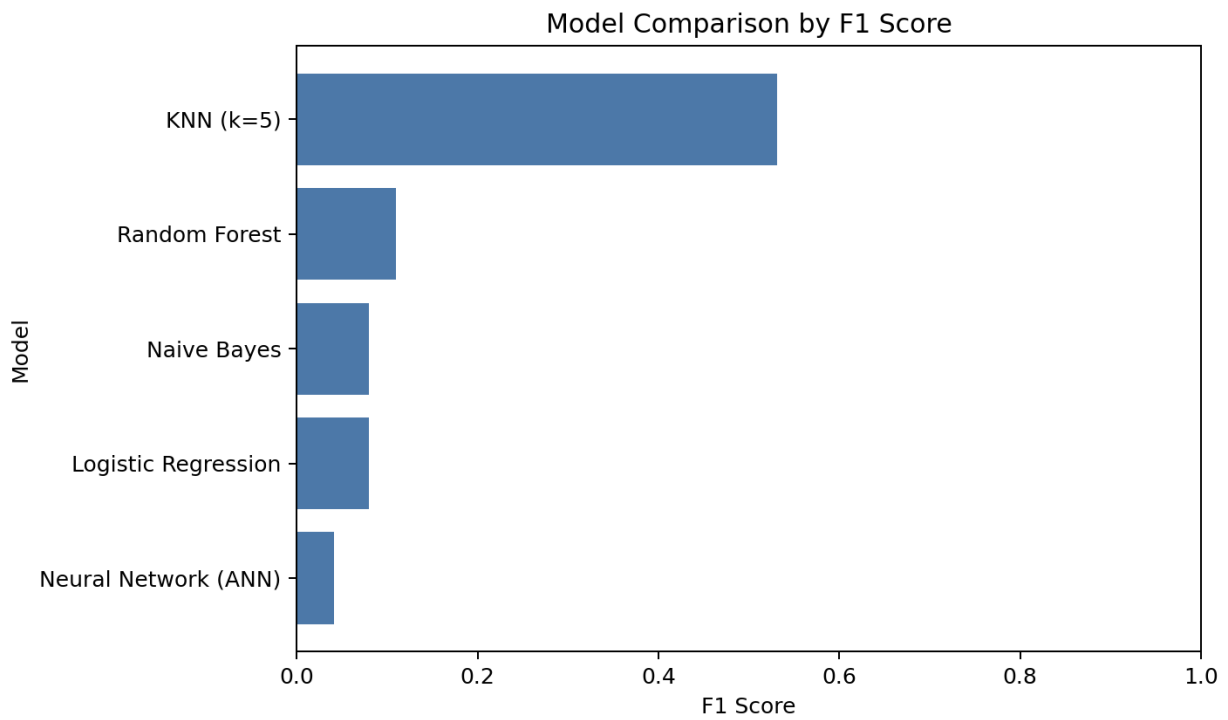


Figure 1: Model comparison table figure

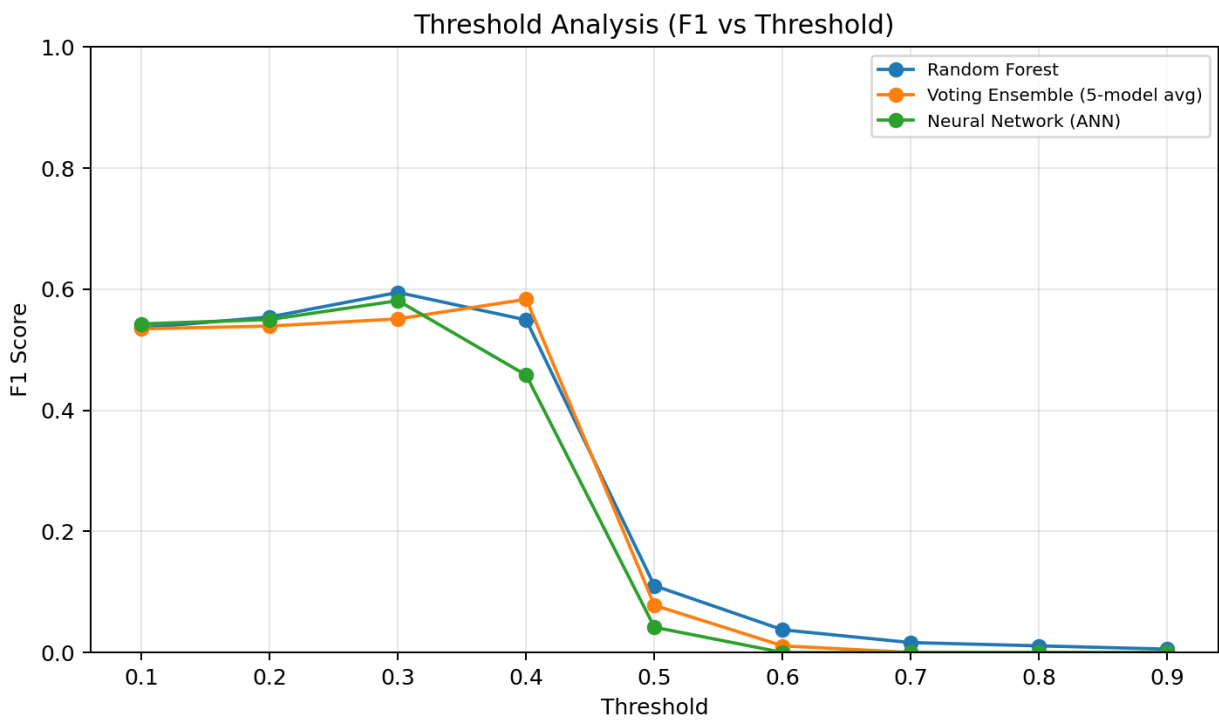


Figure 2: Threshold analysis

5.3 Confusion Matrix Insights

Interpret false positives and false negatives in business terms.

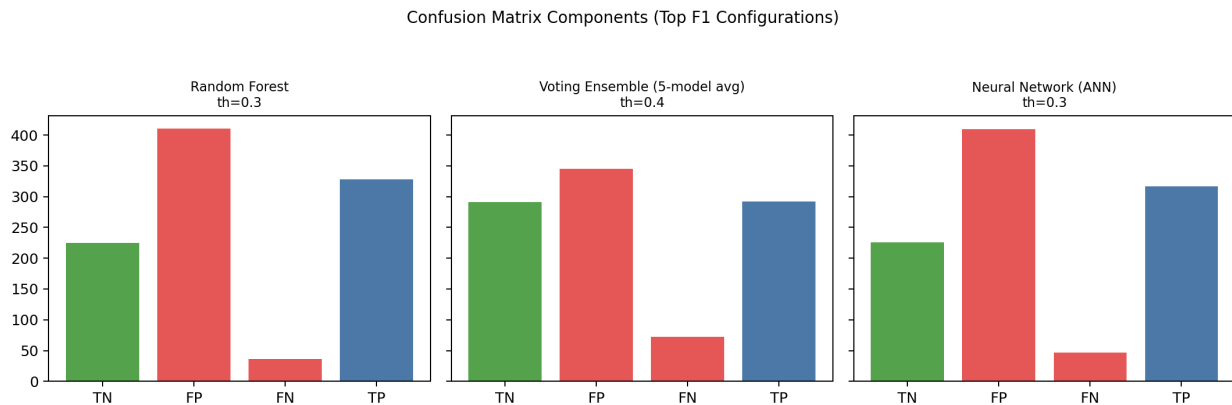


Figure 3: Confusion matrix

5.4 Key Findings

Provide clear, practical conclusions.

6 Conclusion

6.1 Objective Recap

Summarize initial objective.

6.2 Methods Recap

Summarize methods used.

6.3 Main Results

Summarize strongest findings and selected model.

6.4 Business Insights

Summarize actionable recommendations.

7 Appendix (Optional)

7.1 Evolution and Corrections

Reference the mistakes-to-improvements timeline.

7.2 Additional Tables

Add detailed model metrics if needed.